

ENGINEERING CALCULATION NOTE / REVISION B

Spindle vibration reduction

Stepped-beam baseline and rotordynamic sensitivity

Prepared for Jerich Lee

Units: N, m, kg, s (dimensions also shown in mm)

Baseline result under the declared assumptions. A five-section, laterally clamped overhang gives **336.8 Hz by Rayleigh’s method** and **333.7 Hz with two retained stations**. The latter is **8.64% above** the report’s 307.17 Hz ANSYS result. Image-estimated dimensions and idealized support prevent this from being a validated prediction of the real spindle.

The simplest useful calculation

$$\text{geometry + restraint} \longrightarrow k_{\text{eff}}, m_{\text{eff}} \longrightarrow f_1 \simeq \frac{1}{2\pi} \sqrt{\frac{k_{\text{eff}}}{m_{\text{eff}}}}$$

(1)

Retain the changing bending stiffness and weight distributed mass by its motion, rather than putting the whole spindle mass on one spring.

Baseline model / evidence	First frequency	Versus 307.17 Hz
Uniform 60 mm steel cantilever, $L = 410$ mm	255.2 Hz	−16.93%
Stepped overhang, one Rayleigh coordinate	336.8 Hz	+9.64%
Stepped overhang, two-station reduction	333.7 Hz	+8.64%
Original ANSYS result, Figure 18 / Table 5 ^[1]	307.17 Hz	Reference

Added in Revision B: quantify the omitted gyroscopic effect

A separate one-shape, two-plane extension adds cross-sectional rotary inertia and gyroscopic coupling. Rotary inertia lowers the zero-spin result to **334.43 Hz**. Assuming all five equivalent sections rotate at 3000 rpm gives **333.73 and 335.13 Hz**: approximately ±0.208% **relative to that extended zero-spin result**. The original bending calculations remain unchanged.

What is retained, and what remains outside the model

The local beam retains the motor-side root, overhang, gear, stepped tail, and distributed mass. The baseline is nonrotating; the added diagnostic checks gyroscopy only. Tool-side motion, housing flexibility, coaxial internals, verified bearing stiffness, centrifugal prestress, physical damping, and cutting loads remain unresolved.

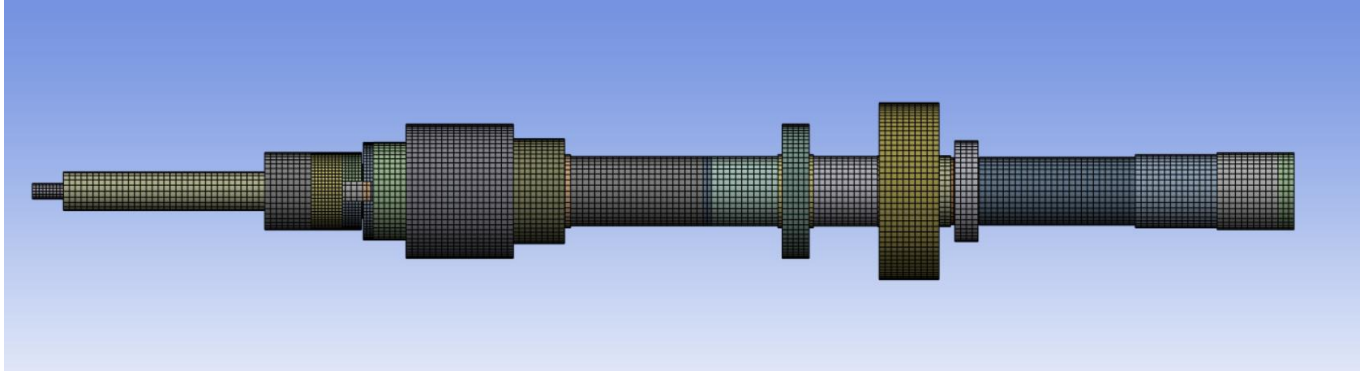
Not a reconstruction of the original solver. Geometry is image-informed, not CAD-verified or calibrated to 307/383 Hz. The new all-five-rotating assumption differs from the report’s exclusion of the rotary union from assigned spin.^[1, p. 28] The ±0.208% result supports a baseline simplification within this surrogate; it is not a bound on real-machine rotational effects.

Reading route: physical model and inputs, pp. 2–3; original derivation, pp. 4–8; FE comparison and bearing change, pp. 9–10; sensitivities/supports, pp. 11–12; new rotordynamics, pp. 13–18; interview and limits, pp. 19–20; checks and references, pp. 21–24.

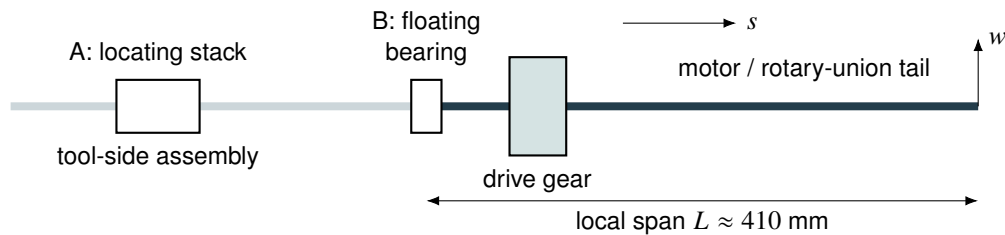
01 / PHYSICAL MODEL

Model the overhang that actually bends

The report's first ANSYS mode is concentrated at the motor-end tail, opposite the cutting tool. Its assembly has four locating bearings near the tool and a floating radial bearing closer to the drive gear. The rotary-union region extends beyond that floating bearing.^[1, pp. 14–15, 31]



Original report Figure 13, PDF p. 28 (printed p. 27). Undeformed mesh; the thin flange before the large gear is the floating bearing. The smaller ring after the gear is not that bearing.



Why a lateral clamp is used at B

The report uses grounded outer bearing faces and *deformable cylindrical kinematic joints* at the shaft interfaces.^[1, pp. 27–29] A cylindrical joint permits axial translation and axial spin, but restrains the two transverse translations and two transverse rotations.^[3] Thus an ideal radial pin is not automatically equivalent to the recorded FE joint.

For this local reduction, take the *limiting* restraint

$$w(0) = 0, \quad w'(0) = 0. \quad (2)$$

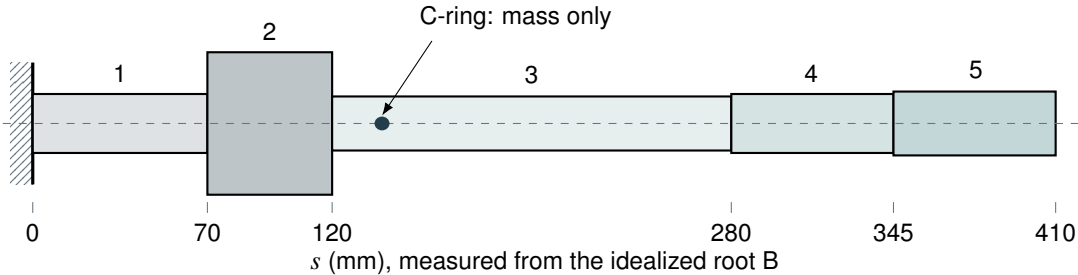
This condenses the tool-side assembly and bearing attachment into a rigid lateral root. It is not an exact replacement for the three-dimensional deformable coupling, bearing-body deformation, or actual rolling contacts. In ANSYS, “deformable” describes how the attachment geometry can deform; it does not itself supply a measured rolling-bearing stiffness.^[4]

Spin and bending slope are different rotations. Restraining $w'(0)$ does not mean welding the spindle against rotation about its own axis. Likewise, “floating” means axial freedom in this assembly, not absence of radial support. Axial motion and spin are outside the baseline one-plane bending model.

The source report uses several global-axis conventions in its illustrations. Here s always means distance along the local shaft axis and w is one transverse displacement.

02 / INPUTS

A five-section approximation, not a CAD audit



Region	Assumed equivalent body	s (mm)	D (mm)	E (GPa)	ρ (kg/m ³)
1	Steel neck	0–70	60	205	7850
2	Bonded drive-gear envelope	70–120	145	210	7800
3	Aluminum tail stem	120–280	55	71	2770
4	Aluminum rear step	280–345	60	71	2770
5	Aluminum terminal step	345–410	65	71	2770

E and ρ : report Table 2. The report assigns steel to the shaft and gear, and aluminum to the rotary union/drawbar. Mapping the visible tail to regions 3–5, and treating the gear as one bonded section, are analyst assumptions rather than verified CAD assignments.^[1, pp. 25–27]

How the dimensions were estimated

Figure 18's 400 mm scale bar spans approximately 502 pixels in the extracted image. Matching the gear width between Figures 18 and 13 transfers an approximate scale of 1.5 pixels/mm to the undeformed mesh image. There, B is near pixel 976 and the tail end near 1590: $(1590 - 976)/1.5 \approx 409$ mm. The gear edges and tail shoulders give rounded stations near 70, 120, 280, and 345 mm. Diameter estimates come from the *undeformed* silhouette, not the exaggerated modal deflection.^[1, Figs. 13, 18]

Section and accessory assumptions

Use solid equivalent sections, $d = 0$:

$$A = \frac{\pi D^2}{4}, \quad I = \frac{\pi D^4}{64}, \quad \mu = \rho A. \quad (3)$$

Represent the thin C-ring by extra mass at $s_r = 140$ mm, with assumed OD/ID/width of 80/55/20 mm:

$$m_r = 7800 \frac{\pi}{4} (0.080^2 - 0.055^2) (0.020) = 0.4135 \text{ kg}. \quad (4)$$

Visible envelopes do not reveal the internal load path. No bore dimensions, drawbar clearances, bearing contact stiffnesses, or rotary-union interface stiffnesses are established. The solid gear includes its core once; the C-ring adds only annular mass. Coaxial internals are not separately added. These assumptions affect both stiffness and inertia and do not produce a rigorous bound.

03 / FIRST PASS

Start with the cantilever you can derive

The model you may be thinking of is the **Jeffcott rotor**: a simple shaft-and-disk idealization. Its familiar centered-disk form is useful for whirl and imbalance, but does not directly preserve this assembly's long, stepped overhang and distributed tail motion.^[6] Here the simplest starting point is a uniform Euler–Bernoulli cantilever, followed by a stepped refinement.

1. Write the equation and boundary conditions

For a uniform beam without rotation, damping, or applied load,

$$EI \frac{\partial^4 w}{\partial s^4} + \rho A \frac{\partial^2 w}{\partial t^2} = 0. \quad (5)$$

With $w(s, t) = Y(s) \cos \omega t$, the root is laterally clamped and the tail is free:

$$Y(0) = Y'(0) = 0, \quad Y''(L) = Y'''(L) = 0. \quad (6)$$

The free-end conditions are zero bending moment and zero shear force. Substitution gives

$$Y'''' - \left(\frac{\beta}{L}\right)^4 Y = 0, \quad \cosh \beta \cos \beta = -1. \quad (7)$$

Its first roots are $\beta_1 = 1.87510$ and $\beta_2 = 4.69409$, so

$$f_n = \frac{\beta_n^2}{2\pi L^2} \sqrt{\frac{EI}{\rho A}}. \quad (8)$$

2. Substitute a 60 mm steel envelope

For $L = 0.410$ m, $D = 0.060$ m, $E = 205 \times 10^9$ Pa, and $\rho = 7850$ kg/m³,

$$A = 2.82743 \times 10^{-3} \text{ m}^2, \quad I = 6.36173 \times 10^{-7} \text{ m}^4, \quad (9)$$

$$f_1 = \frac{1.87510^2}{2\pi(0.410)^2} \sqrt{\frac{(205 \times 10^9)(6.36173 \times 10^{-7})}{7850(2.82743 \times 10^{-3})}} = \boxed{255.2 \text{ Hz}}, \quad (10)$$

$$f_2 = \boxed{1599.1 \text{ Hz}}. \quad (11)$$

These are modes of the *uniform surrogate*, not identified modes of the complete spindle. Its first estimate is 16.93% below 307.17 Hz.

3. Learn the design scaling before adding detail

Because $I/A = D^2/16$ for a solid circle,

$$f_n \propto \frac{D}{L^2} \sqrt{\frac{E}{\rho}}. \quad (12)$$

Shorter overhangs and larger diameters tend to raise bending frequencies. A stronger steel grade does not necessarily have a much larger E/ρ . This helps explain why changing steel grades had little effect in the original report.^[1, Table 5]

Why refine this result? The stepped approximation has a wide, stiff gear near the root and a lighter aluminum tail with several diameter changes. One average diameter and one material cannot retain that distribution. The next calculation changes the distribution explicitly, rather than tuning the uniform beam to 307 Hz.

04 / STATIC FLEXIBILITY

Turn the stepped overhang into a spring

Apply a small conceptual transverse force P at $s = L$. This is a mathematical probe to obtain flexibility, not a claimed milling load. For the cantilever,

$$M(s) = P(L - s), \quad U = \frac{1}{2} \int_0^L \frac{P^2(L - s)^2}{E(s)I(s)} ds. \quad (13)$$

From $\delta_L = \partial U / \partial P$, the tail compliance and equivalent stiffness are

$$C_{LL} = \frac{\delta_L}{P} = \sum_j \frac{(L - a_j)^3 - (L - b_j)^3}{3E_j I_j}, \quad k_L = C_{LL}^{-1}. \quad (14)$$

Here a_j, b_j are the section endpoints in *meters*. Different sections have the same moment diagram but different curvatures $M/(EI)$.

A complete worked term: the 60 mm root neck

$$I_1 = \frac{\pi(0.060)^4}{64} = 6.36173 \times 10^{-7} \text{ m}^4, \quad (15)$$

$$C_1 = \frac{(0.410 - 0)^3 - (0.410 - 0.070)^3}{3(205 \times 10^9)(6.36173 \times 10^{-7})} = 7.56992 \times 10^{-8} \text{ m/N}. \quad (16)$$

Region	Body	$I \text{ (mm}^4\text{)}$	$EI \text{ (N m}^2\text{)}$	$C_j \text{ (10}^{-7} \text{ m/N)}$
1	Steel neck	636,173	130,415	0.75699
2	Gear envelope	21,699,109	4,556,813	0.01091
3	Tail stem	449,180	31,892	2.31951
4	Rear step	636,173	45,168	0.14187
5	Terminal step	876,241	62,213	0.01471
Total				3.24399

$$C_{LL} = 3.24399 \times 10^{-7} \text{ m/N}, \quad k_L = 3.08262 \times 10^6 \text{ N/m}. \quad (17)$$

As a units check, an illustrative $P = 100 \text{ N}$ at this motor-end tail gives

$$\delta_L = PC_{LL} = 3.24399 \times 10^{-5} \text{ m} = 32.44 \text{ } \mu\text{m}. \quad (18)$$

That is not the cutting-tool displacement: the force is applied at a different location, and the tool-side geometry is absent.

The weak span matters more than the largest visible part. The 160 mm aluminum stem contributes about 71.5% of this static bending compliance; the steel root neck contributes 23.3%. The wide gear contributes only 0.34%. These percentages describe this assumed beam, not extracted ANSYS strain-energy fractions.

05 / RAYLEIGH ESTIMATE

Use the mass that participates in the mode

Approximate the first mode by the static tip-load deflection $v(s)$ and normalize it as $\phi(s) = v(s)/v(L)$. This is a Rayleigh energy approximation; no vibration amplitude needs to be assumed.^[7]

Build the shape one section at a time

Within section j , let $u = s - a_j$. Integrating $v'' = P(L - s)/(E_j I_j)$ gives

$$v(s) = v(a_j) + v'(a_j)u + \frac{P}{E_j I_j} \left[\frac{(L - a_j)u^2}{2} - \frac{u^3}{6} \right], \quad (19)$$

$$v'(s) = v'(a_j) + \frac{P}{E_j I_j} \left[(L - a_j)u - \frac{u^2}{2} \right]. \quad (20)$$

Start from $v(0) = v'(0) = 0$ and carry the endpoint displacement and slope into the next section. The resulting shape is continuous in displacement and slope; curvature can change where EI changes.

Reduce the kinetic and strain energies

Set $w(s, t) = \phi(s)q(t)$ with $\phi(L) = 1$. Then

$$T = \frac{1}{2} \left[\underbrace{\sum_j \rho_j A_j \int_{a_j}^{b_j} \phi^2 ds + m_r \phi(s_r)^2}_{m_{\text{eff}}} \right] \dot{q}^2, \quad (21)$$

$$U = \frac{1}{2} k_L q^2, \quad m_{\text{eff}} \ddot{q} + k_L q = 0. \quad (22)$$

Region	Body	Mass (kg)	ϕ at right end	Modal mass (kg)
1	Steel neck	1.55367	0.02239	0.000159
2	Gear envelope	6.44007	0.05369	0.009823
3	Tail stem	1.05297	0.44748	0.059705
4	Rear step	0.50908	0.71522	0.173719
5	Terminal step	0.59746	1.00000	0.442895
–	C-ring extra mass	0.41351	–	0.002131
Total		10.56676		0.688433

Modal mass uses the integral of ϕ^2 , not each section's endpoint value times its mass. The shape is cubic within each material interval; its squared polynomial can be integrated directly.

$$f_R = \frac{1}{2\pi} \sqrt{\frac{3.08262 \times 10^6}{0.688433}} = 336.8 \text{ Hz.} \quad (23)$$

A large component need not dominate a mode's inertia. The 6.44 kg gear moves little in this shape and contributes only about 0.0098 kg to tip-normalized modal mass. The 0.60 kg terminal step contributes about 0.44 kg. Using all 10.57 kg as a tip mass would describe the wrong motion.

06 / TWO-COORDINATE MODEL

Obtain two modes without a large FE model

Retain two physical displacements: gear-center motion $q_1 = w(s_g)$ at $s_g = 0.095$ m and tail motion $q_2 = w(L)$ at $L = 0.410$ m. This is a **two-station Rayleigh–Ritz reduction**, not two guessed lumped masses.

Static flexibility provides the interpolation

The displacement at s due to a unit transverse force at another axial station ξ is

$$H(s, \xi) = \int_0^{\min(s, \xi)} \frac{(s-u)(\xi-u)}{E(u)I(u)} du. \quad (24)$$

Each section contributes, with $z = \min(s, \xi, b_j) > a_j$,

$$H_j(s, \xi) = \frac{s\xi(z - a_j) - (s + \xi)(z^2 - a_j^2)/2 + (z^3 - a_j^3)/3}{E_j I_j}. \quad (25)$$

Let $s_1 = s_g, s_2 = L$ and $C_{ij} = H(s_i, s_j)$. Under forces $\mathbf{P}$, $\mathbf{q} = \mathbf{C}\mathbf{P}$. Hence the reduced stiffness is $\mathbf{K} = \mathbf{C}^{-1}$, and the reconstructed beam shape is

$$w(s, t) = \mathbf{N}(s)\mathbf{q}(t), \quad \mathbf{N}(s) = \begin{bmatrix} H(s, s_g) & H(s, L) \end{bmatrix} \mathbf{C}^{-1}. \quad (26)$$

The interpolation exactly returns $\mathbf{N}(s_g) = [1 \ 0]$ and $\mathbf{N}(L) = [0 \ 1]$.

Keep the continuous mass distribution

$$\mathbf{M} = \sum_j \rho_j A_j \int_{a_j}^{b_j} \mathbf{N}^T \mathbf{N} ds + m_r \mathbf{N}(s_r)^T \mathbf{N}(s_r). \quad (27)$$

The entries follow by integrating products of piecewise cubic functions, just as on the previous page. Stiffness can also be checked from $\int E I N''^T N'' ds$.

$$\mathbf{C} = 10^{-7} \begin{bmatrix} 0.0215260 & 0.1231871 \\ 0.1231871 & 3.2439936 \end{bmatrix} \text{ m/N}, \quad (28)$$

$$\mathbf{K} = 10^6 \begin{bmatrix} 593.53793 & -22.53896 \\ -22.53896 & 3.938513 \end{bmatrix} \text{ N/m}, \quad (29)$$

$$\mathbf{M} = \begin{bmatrix} 12.188384 & 0.788953 \\ 0.788953 & 0.610937 \end{bmatrix} \text{ kg}. \quad (30)$$

M_{11} is not the physical gear mass, nor is M_{22} the physical tail mass. They are generalized inertia coefficients for the selected displacement coordinates. The off-diagonal term is retained because the interpolated motions share distributed material.

Rayleigh–Ritz trial shapes must satisfy the root kinematic constraints. They need not satisfy the unloaded free-end shear condition exactly: natural boundary conditions are approached through the energy approximation as the trial space is enriched.

07 / NATURAL FREQUENCIES AND SHAPES

Solve a quadratic, then inspect the motion

Free vibration satisfies $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0}$. With $\mathbf{q} = \mathbf{a} \cos \omega t$,

$$\det(\mathbf{K} - \lambda \mathbf{M}) = 0, \quad \lambda = \omega^2. \quad (31)$$

For symmetric 2×2 matrices this becomes

$$0 = (M_{11}M_{22} - M_{12}^2)\lambda^2 - (K_{11}M_{22} + K_{22}M_{11} - 2K_{12}M_{12})\lambda + (K_{11}K_{22} - K_{12}^2). \quad (32)$$

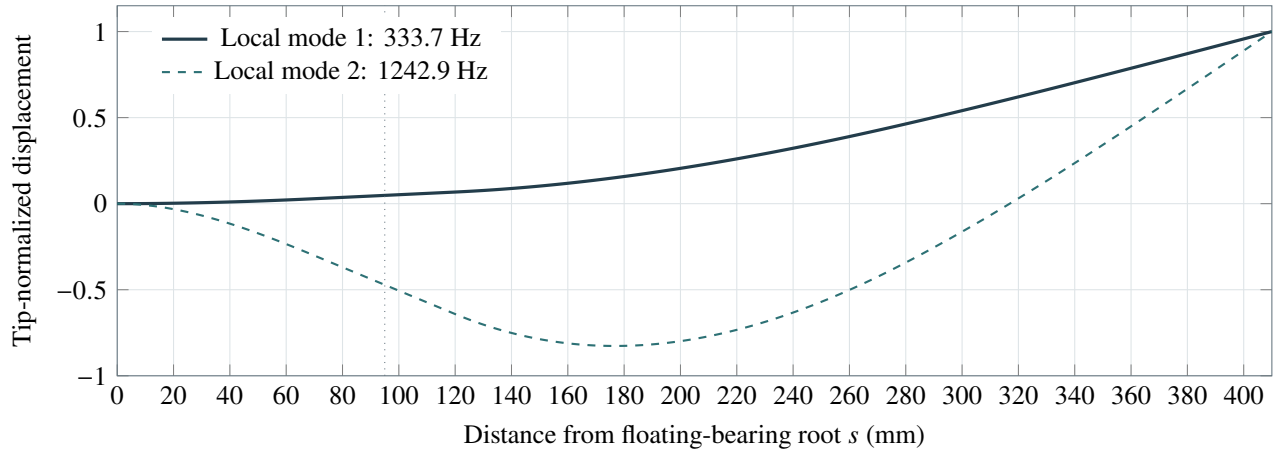
Using $x = \lambda/(10^6 \text{ s}^{-2})$ and dividing by a common dimensional factor gives the numerical quadratic

$$6.823892x^2 - 446.182993x + 1829.651991 = 0. \quad (33)$$

Thus $x_1 = 4.396264$, $x_2 = 60.989148$, and

$$f_{1,\text{local}} = 333.7 \text{ Hz}, \quad f_{2,\text{local}} = 1242.9 \text{ Hz}. \quad (34)$$

For unit tail displacement, the coordinate vectors are $\mathbf{a}_1 = [0.04817 \ 1]^\top$ and $\mathbf{a}_2 = [-0.47161 \ 1]^\top$. Reconstruct each shape from $\psi_n(s) = \mathbf{N}(s)\mathbf{a}_n$.

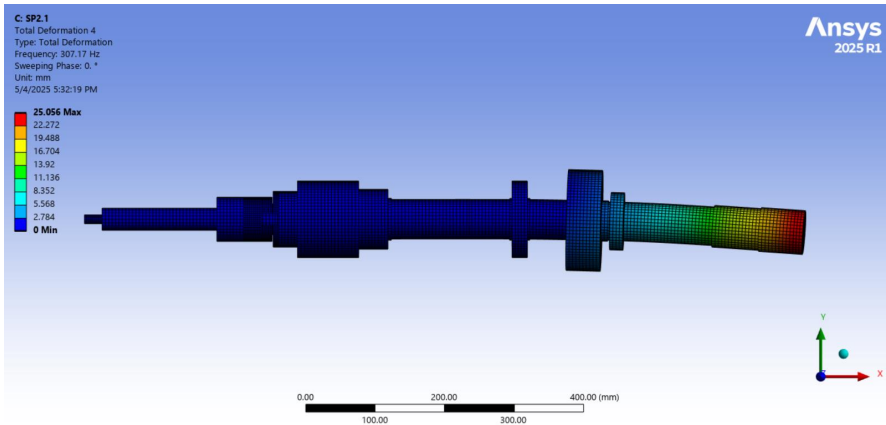


Computed shapes of the assumed overhang, not measured or digitized ANSYS eigenvectors. The dotted line marks the gear center. Mode 2 has an interior zero near $s = 317$ mm and opposing gear/tail motion.

The second local mode is not the report's second system mode. Table 5 lists 523.67 Hz, but supplies no corresponding mode shape. The omitted tool, inter-bearing span, joints, or another motion family can produce modes below 1243 Hz. No matched-mode error is assigned to that pair. In the nonrotating circular idealization, each local bending family has an equal-frequency orthogonal partner. The separate one-shape extension on page 15 illustrates gyroscopic splitting of a lateral pair.

08 / COMPARISON WITH THE ORIGINAL MODEL

Agreement in scale, not a validation claim



Original Figure 18, PDF p. 32 (printed p. 31), unmodified. The screenshot reads 307.17 Hz. Its maximum deformation legend is eigenvector scaling, not measured vibration amplitude.^[1,5]

Define the comparison before calling it an “error”

Use the FE readout as a reference, not as experimental truth:

$$e_{\text{HC|FE}} = 100 \frac{f_{\text{HC}} - f_{\text{FE}}}{f_{\text{FE}}}.$$

(35)

$$e_R = 100 \frac{336.7824 - 307.17}{307.17} = +9.64\%,$$

(36)

$$e_{2q} = 100 \frac{333.7045 - 307.17}{307.17} = \boxed{+8.64\%}.$$

(37)

The two-coordinate result is 26.53 Hz higher. Its first shape is qualitatively tail-dominated, like the shown FE mode. The report does not provide nodal eigenvectors for a quantitative shape correlation such as a modal assurance criterion.

Keep three distinct checks separate

Check	What the supplied record supports
Hand model versus FE	About 9% frequency discrepancy, but geometry, support representation, rotational treatment, and retained subassembly differ.
FE mesh sensitivity	Reported first frequencies of 305–307 Hz across four meshes. This is mesh sensitivity, not a physical error bound. ^[1, Table 3]
FE versus observation	An operational band near 310 Hz; 307.17 Hz is 0.91% lower. Impact tests were inconclusive, so a measured full mode shape is not established by the report. ^[1, pp. 19, 22, 30]

The report calls the model “validated.” Here the evidence supports **frequency correlation**, not a measured full mode shape or proof of loaded cutting stability. The new hand calculation is a separate, assumption-dependent comparison.

09 / BEARING MODIFICATION

Separate a support shift from a new bearing stack

The report associates the recommended modification with moving the floating stack 22 mm toward the drive and a first-mode shift from approximately 307 to 383 Hz. Figure 21 reads 383.27 Hz. [1, pp. 33–34]

A controlled hand-calculation sensitivity

Move only the idealized root by $\Delta = 22$ mm, leaving the physical gear and tail geometry unchanged. In the new coordinate $s' = s - \Delta$,

$$L' = 0.410 - 0.022 = 0.388 \text{ m}, \quad s'_g = 0.073 \text{ m}. \quad (38)$$

The first section shortens from 70 to 48 mm. The remaining section lengths do not change; the C-ring moves to $s'_r = 118$ mm in the new coordinate. Repeating exactly the same calculations gives

Quantity	Original root	Root shifted 22 mm
Tip compliance (10^{-7} m/N)	3.24399	2.97537
Tip stiffness (10^6 N/m)	3.08262	3.36093
Rayleigh effective mass (kg)	0.68843	0.65662
Rayleigh frequency (Hz)	336.78	360.07
Two-coordinate first frequency (Hz)	333.70	358.30

$$\frac{358.3014}{333.7045} - 1 = 7.37\% \quad \text{versus} \quad \frac{383.27}{307.17} - 1 = 24.77\% \quad (\text{reported FE}). \quad (39)$$

What this comparison says

The translation-only model predicts the expected upward trend, but does *not* reproduce the reported 24.8% gain. Its already-clamped root leaves no additional root tilt to suppress. Adding and spacing real bearings, or changing FE joint scopes, can alter effective restraint beyond merely shifting one point. Their contribution cannot be quantified from the available geometry and setup.

Although 358.30 Hz is only 6.51% below 383.27 Hz, that scalar proximity is **not** agreement for the actual redesign: the hand case and FE modification are different cases.

A useful inverse check, explicitly not a prediction

For unchanged effective mass, the reported frequency ratio would require

$$\frac{k_{\text{new}}}{k_{\text{old}}} = \left(\frac{383.27}{307.17} \right)^2 = 1.55687, \quad (40)$$

or about 55.7% more effective stiffness. This is inferred from the frequencies, not independently calculated bearing stiffness.

The bearing count is inconsistent within the report. Section 7.8 and the synopsis describe adding one bearing (two total); the following page and Figure 21 say two additional bearings. Table 4 labels the 383 Hz case “Modified 2x.” This note preserves that label and does not silently decide which physical stack was simulated. Confirm the original CAD/joint scopes before reconstructing it.

10 / SENSITIVITY

The input uncertainty is larger than the arithmetic

The decimal places in the calculation allow reproduction; they are not dimensional accuracy or a statistical confidence interval. The image estimates, internal geometry, and support model deserve more scrutiny than the last hertz.

Geometric scaling within the original nonrotating baseline

With unchanged materials, scaling all diameters by γ gives $EI \propto \gamma^4$, mass per length $\propto \gamma^2$, and $f \propto \gamma$. Scale the C-ring's OD and ID with the other diameters. Scaling every axial length by α (including ring width and station positions) gives $f \propto \alpha^{-2}$. Scaling *all dimensions* by α gives $f \propto \alpha^{-1}$.

Assumed change from the baseline	First mode (Hz)	Change
All axial lengths -5% ; diameters fixed	369.8	+10.80%
All axial lengths $+5\%$; diameters fixed	302.7	-9.30%
All diameters -5% ; axial lengths fixed	317.0	-5.00%
All diameters $+5\%$; axial lengths fixed	350.4	+5.00%
All dimensions -5% together	351.3	+5.26%
All dimensions $+5\%$ together	317.8	-4.76%

These are chosen perturbations, not measured image-error bounds. A common image-scale error changes lengths and diameters together; it is not the same scenario as changing axial lengths alone.

Hollow and coaxial regions need both stiffness and mass changes

For an actual tube,

$$A = \frac{\pi}{4}(D^2 - d^2), \quad I = \frac{\pi}{64}(D^4 - d^4), \quad \frac{I}{A} = \frac{D^2 + d^2}{16}. \quad (41)$$

Removing material lowers both EI and mass. A uniform tube with the same outer diameter can have a *higher* self-mass bending frequency than a solid beam, even though its static stiffness is lower. Therefore “assuming solid must overpredict frequency” is not generally valid. A separate tip mass or coaxial contact arrangement changes the conclusion again.

Refinement order

First replace the image geometry with CAD section dimensions and component masses. Next verify the bending restraint and rotary-union/shaft connection. Revision B adds a fixed-shape rotary-inertia and gyroscopic check (pages 13–18); shear deformation, actual spin partition, and centrifugal-prestress effects remain open. The equivalent $L/D \approx 6.8$ and short, wide gear limit the fidelity of Euler–Bernoulli higher modes.

Do not tune the length until the answer becomes 307 Hz. Keep independently obtained geometry fixed, calculate the discrepancy, then test physical hypotheses such as support compliance or a different internal load path. Matching one frequency cannot uniquely identify all of these inputs.

11 / BOUNDARY-CONDITION EXTENSION

What changes when the bearing can move or tilt?

For a real assembly, replace the ideal root with a translational spring k_y and bending-rotation spring k_θ . Assume these springs are massless, linear, uncoupled, and referred to ground. The tip compliance becomes

$$C_{LL,\text{total}} = C_{LL,\text{beam}} + \frac{1}{k_y} + \frac{L^2}{k_\theta}. \quad (42)$$

A force P causes root translation P/k_y and moment PL , giving slope PL/k_θ and extra tip motion PL^2/k_θ .

The same two-coordinate method still works

Replace the influence function on page 7 by

$$H_{\text{total}}(s, \xi) = H_{\text{beam}}(s, \xi) + \frac{1}{k_y} + \frac{s\xi}{k_\theta}. \quad (43)$$

Rebuild $\mathbf{C}$, $\mathbf{N}$, and $\mathbf{M}$, not just $\mathbf{K}$: support motion changes the modal mass distribution too. A more general root compliance matrix can include translation–rotation coupling. No numerical bearing stiffness is assigned here because none is documented.

Why two radial pins are not a clamp

For a *separate uniform-beam example*, place radial supports at $A = -a$ and $B = 0$, with a tail load P at L . Static equilibrium gives

$$R_A = -\frac{PL}{a}, \quad R_B = P \left(1 + \frac{L}{a}\right). \quad (44)$$

With constant EI , free tilt at each support, and radial stiffnesses k_A, k_B , energy in both spans and supports gives

$$C = \frac{L^2(a+L)}{3EI} + \frac{(L/a)^2}{k_A} + \frac{(1+L/a)^2}{k_B}. \quad (45)$$

Even for $k_A, k_B \rightarrow \infty$, the inter-bearing span supplies extra flexibility $L^2a/(3EI)$ compared with a local clamp. This is a different restraint from the primary calculation; it is not an extra correction to add blindly to an already condensed assembly.

How spacing a bearing pair can resist tilt

For two identical radial springs, each stiffness k , separated axially by d , their displacements under a small rigid-section translation y and tilt θ are $y \mp d\theta/2$. Therefore

$$U = \frac{1}{2}k(y - d\theta/2)^2 + \frac{1}{2}k(y + d\theta/2)^2 \quad (46)$$

$$= \frac{1}{2}(2k)y^2 + \frac{1}{2}\left(\frac{kd^2}{2}\right)\theta^2, \quad k_y = 2k, \quad k_\theta = \frac{kd^2}{2}. \quad (47)$$

Bearing spacing can matter as much as bearing count. A pair supplies a moment-resisting lever arm, while a single ideal radial point spring does not. Axial thermal freedom must still be preserved in the actual arrangement. This mechanism is explanatory; it is not a reconstruction of the ambiguous “Modified 2x” FE case.

12 / ROTORDYNAMIC SENSITIVITY

Keep the bending baseline; check the omission

The original calculation asks whether the motor-end overhang's stiffness, mass distribution, and restraint explain a bending frequency of a few hundred hertz. Its one-plane equation,

$$m_t \ddot{q} + kq = 0, \quad f_R = \frac{1}{2\pi} \sqrt{\frac{k}{m_t}}, \quad (48)$$

is a useful baseline, but is not an operating-speed rotor model. Here m_t is the original tip-normalized translational mass m_{eff} , and $k = k_L$.

The smallest extension that represents gyroscopic coupling

A spinning cross-section carries angular momentum. Tilting its axis during bending produces a gyroscopic moment that couples the two transverse directions. A stationary-frame beam formulation includes both cross-sectional rotary inertia and gyroscopic coupling.^[9]

Retain exactly the static shape $\phi(s)$ from page 6, but let it move in two perpendicular, fixed lateral planes:

$$w_x(s, t) = \phi(s)q_x(t), \quad w_y(s, t) = \phi(s)q_y(t), \quad \phi(L) = 1. \quad (49)$$

The baseline kinematic restraint is unchanged: $\phi(0) = \phi'(0) = 0$. The two generalized coordinates are the lateral tip displacements q_x, q_y , measured in meters.

Item	Declared choice for this diagnostic
Geometry and materials	Same five solid circular equivalents and report-based properties; no new dimensions or fitted parameters.
Spatial approximation	One frozen static-deflection shape, identical in both planes; small deflection and Euler–Bernoulli kinematics.
Section spin	All five equivalent sections co-rotate at a prescribed, constant 3000 rpm. This is a <i>new assumption</i> .
C-ring	Original extra translational point mass retained; its diametral and polar inertias are not added.
Stiffness and losses	Same k as the baseline; no centrifugal-prestress calculation, shear correction, physical damping, or cutting force.

The spin assignment does not match the reported ANSYS scope. The report excludes the bearing stacks and rotary union from its 3000 rpm assignment; the physical rotary union also includes a nonrotating end.^[1, pp. 15, 28] The equivalent aluminum tail does not resolve those internal components. Thus the all-five-rotating case is a reduced-model sensitivity, not a reconstruction or verified conservative bound for the real spindle.

Two planes is not the same as two stations. Pages 7–8 retain gear and tail displacements in *one* plane to obtain two spatial modes. This extension retains *one* spatial shape in two planes to obtain a whirl pair. The earlier gear station is written s_g ; g below denotes a gyroscopic coefficient.

13 / ROTARY INERTIA

Integrate the slope, not just the displacement

For tip-normalized ϕ , translating material is weighted by ϕ^2 , while cross-sectional tilting is weighted by $(\phi')^2$. Define

$$m_t = \sum_j \rho_j A_j \int_{a_j}^{b_j} \phi^2 ds + m_r \phi(s_r)^2 = 0.68843256 \text{ kg}, \quad (50)$$

$$m_\theta = \sum_j \rho_j I_j \int_{a_j}^{b_j} (\phi')^2 ds, \quad g = \sum_j \rho_j J_{p,j} \int_{a_j}^{b_j} (\phi')^2 ds. \quad (51)$$

Here I_j and $J_{p,j}$ are *area* moments, not lumped mass moments. For each circular equivalent, $J_{p,j} = 2I_j$. The cross-sectional inertia terms follow the beam-rotor kinetic-energy formulation; the integration below uses this note's assumed geometry.^[9]

A hand-integrable polynomial for every section

With $u = s - a_j$ and $\ell_j = b_j - a_j$, differentiate the static shape on page 6:

$$\phi'(s) = c_{0j} + c_{1j}u + c_{2j}u^2, \quad c_{0j} = \frac{v'(a_j)}{v(L)}, \quad c_{1j} = \frac{P(L - a_j)}{E_j I_j v(L)}, \quad c_{2j} = -\frac{P}{2E_j I_j v(L)}. \quad (52)$$

The probe force P cancels because $v(L)$ is proportional to P . Squaring and integrating gives

$$\begin{aligned} S_j &= \int_{a_j}^{b_j} (\phi')^2 ds \\ &= c_{0j}^2 \ell_j + c_{0j} c_{1j} \ell_j^2 + \frac{c_{1j}^2 + 2c_{0j} c_{2j}}{3} \ell_j^3 + \frac{c_{1j} c_{2j}}{2} \ell_j^4 + \frac{c_{2j}^2}{5} \ell_j^5. \end{aligned} \quad (53)$$

Thus $m_{\theta,j} = \rho_j I_j S_j$ and, for the declared co-rotating circular sections, $g_j = 2m_{\theta,j}$.

Region	Body	$S_j \text{ (m}^{-1}\text{)}$	$m_{\theta,j} \text{ (kg)}$	$g_j \text{ (kg)}$
1	Steel neck	0.009410	0.00004699	0.00009399
2	Gear envelope	0.019590	0.00331573	0.00663147
3	Tail stem	1.111195	0.00138258	0.00276516
4	Rear step	1.103899	0.00194529	0.00389058
5	Terminal step	1.247709	0.00302842	0.00605684
Total			0.00971902	0.01943803

For example, region 2 gives $m_{\theta,2} = 7800(2.1699109 \times 10^{-5})(0.019590388) = 0.00331573 \text{ kg}$. Units: $(\text{kg/m}^3)(\text{m}^4)(\text{m}^{-1}) = \text{kg}$. The point-mass C-ring contributes to m_t , but not to this table.

$$m = m_t + m_\theta = 0.69815157 \text{ kg}, \quad g = 2m_\theta = 0.01943803 \text{ kg}. \quad (54)$$

Low translation does not imply negligible gyroscopic contribution. In this shape, the gear contributes only about 1.4% of m_t , but about 34.1% of g . Its large polar inertia is weighted by the local *slope*. The final splitting still depends on the ratio g/m , not on gear mass alone.

14 / TWO-PLANE EQUATIONS

Derive the forward and backward whirl pair

Use right-handed fixed axes (x, y, s) and positive spin Ω about $+s$. One consistent sign convention gives the undamped reduced equations

$$m \begin{bmatrix} \ddot{q}_x \\ \ddot{q}_y \end{bmatrix} + \underbrace{\Omega \begin{bmatrix} 0 & g \\ -g & 0 \end{bmatrix}}_{\mathbf{G}_1} \begin{bmatrix} \dot{q}_x \\ \dot{q}_y \end{bmatrix} + k \begin{bmatrix} q_x \\ q_y \end{bmatrix} = \mathbf{0}. \quad (55)$$

This is the specialization of $\mathbf{M}\ddot{\mathbf{q}} + \Omega\mathbf{G}_1\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0}$ to one isotropic lateral shape. $\mathbf{G}_1$ is per unit spin speed; $\Omega\mathbf{G}_1$ has units kg/s. General rotor equations can contain additional damping and speed-dependent terms.^[10]

1. Solve the harmonic eigenproblem

With $\mathbf{q} = \text{Re}\{\mathbf{a}e^{i\omega t}\}$,

$$\det \begin{bmatrix} k - m\omega^2 & i\omega g\Omega \\ -i\omega g\Omega & k - m\omega^2 \end{bmatrix} = 0 \implies (k - m\omega^2)^2 - (g\Omega\omega)^2 = 0. \quad (56)$$

Writing $\omega_0 = \sqrt{k/m}$ and $a = g\Omega/(2m)$, the two positive roots for $\Omega \geq 0$ are

$$\omega_- = \sqrt{\omega_0^2 + a^2} - a, \quad \omega_+ = \sqrt{\omega_0^2 + a^2} + a. \quad (57)$$

At zero spin both equal ω_0 ; spin splits the otherwise degenerate pair. The two frequencies do not introduce an additional spatial bending mode.

2. Check the physical direction, not just the sign label

For the convention above, $q_x = Q \cos \omega t$, $q_y = Q \sin \omega t$ describes forward circular motion about $+s$. It satisfies $k - m\omega^2 + g\Omega\omega = 0$, so its frequency is ω_+ . Reversing the sine sign gives backward motion and ω_- . Forward/backward refers to whirl in the same/opposite direction as shaft rotation.^[11]

3. Gyroscopic coupling is not energy dissipation

The skew symmetry $\mathbf{G}_1^\top = -\mathbf{G}_1$ gives

$$\dot{\mathbf{q}}^\top (\Omega\mathbf{G}_1) \dot{\mathbf{q}} = 0, \quad \frac{d}{dt} \left[\frac{1}{2} m \dot{\mathbf{q}}^\top \dot{\mathbf{q}} + \frac{1}{2} k \mathbf{q}^\top \mathbf{q} \right] = 0. \quad (58)$$

Thus the ideal gyroscopic term changes the modal motion without dissipating energy. A velocity-dependent matrix is not automatically physical damping.

Section spin and section tilting are separate. A nonspinning section can still contribute to m_θ . With actual section speeds Ω_j , replace $g\Omega$ by $\Gamma = \sum_j g_j \Omega_j$, while retaining all tilting mass. No numerical spin partition is assigned here because the equivalent tail does not resolve the real rotary-union internals. The reported numbers use $\Omega_j = \Omega$ for all five sections.

15 / NUMERICAL ROTATION CHECK

A small gyroscopic shift at 3000 rpm

The recomputed quantities use the unchanged stiffness and shape, not rounded table inputs:

$$\Omega = 3000 \frac{2\pi}{60} = 314.15927 \text{ rad/s}, \quad \frac{g}{m} = 0.02784214, \quad (59)$$

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{3.08262016 \times 10^6}{0.69815157}} = 334.43004 \text{ Hz}, \quad \frac{a}{2\pi} = 0.69605352 \text{ Hz}. \quad (60)$$

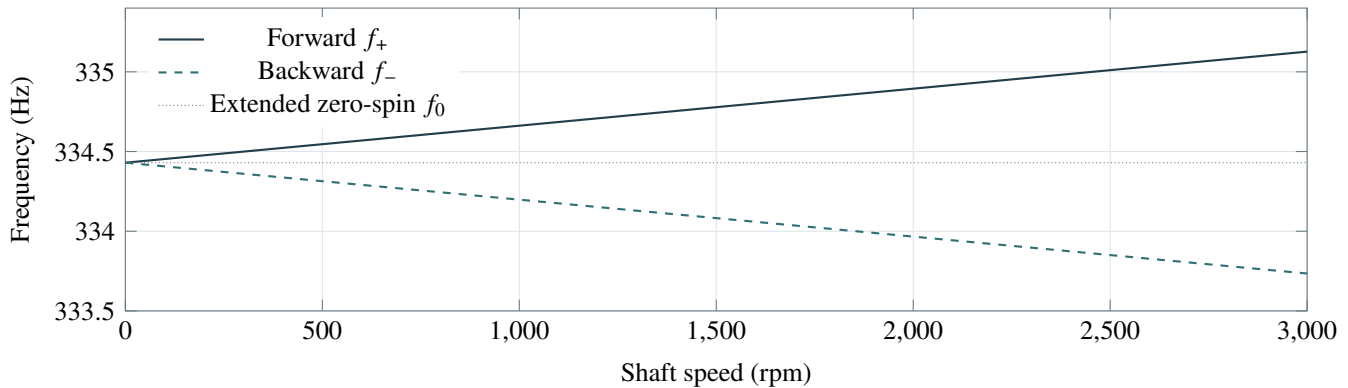
One-shape calculation	Frequency (Hz)	Change / reference
Original: no cross-sectional rotary inertia	336.78244	Baseline
Rotary inertia added, zero spin	334.43004	−0.69849% / baseline
3000 rpm: backward, lower branch	333.73471	−0.20791% / f_0
3000 rpm: forward, upper branch	335.12682	+0.20835% / f_0

Separate the two changes

Adding m_θ lowers the frequency by about 0.70% even at zero spin: cross-sections tilt during vibration whether or not the shaft spins. Turning on spin then produces the approximately $\pm 0.208\%$ gyroscopic correction relative to f_0 , not relative to 336.78 Hz.

$$\varepsilon = \frac{g\Omega}{2m\omega_0} = 0.00208131, \quad \frac{f_{\pm} - f_0}{f_0} = \sqrt{1 + \varepsilon^2} \pm \varepsilon - 1. \quad (61)$$

The two percentage magnitudes differ slightly because of the common positive term $\sqrt{1 + \varepsilon^2} - 1$. The separation is 1.39211 Hz at 3000 rpm.



Calculated branches of the one-shape, all-five-rotating surrogate in the stationary frame. The tight vertical scale reveals the small split; these are not ANSYS results.

Conclusion within the assumed beam: the gyroscopic shift at 3000 rpm is small, supporting its omission from the original baseline explanation. It is not an error bound for the complete spindle or a check of centrifugal prestress. The lower branch's proximity to the separate two-station Ritz result (333.70 Hz) is coincidental; they are different reductions.

16 / EXCITATION AND SPEED

Small frequency shift does not mean small forcing

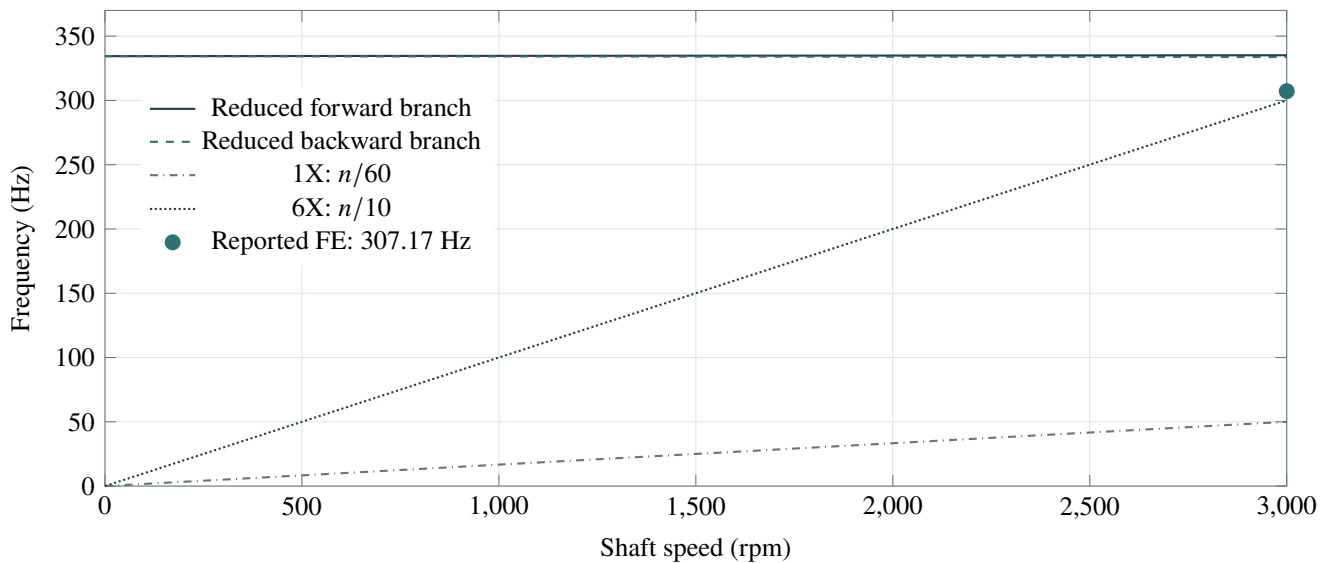
At the report's maximum 3000 rpm, shaft rotation is 50 Hz. With six cutting flutes, the tooth-passing fundamental is 300 Hz.^[1, pp. 8–9] These are excitation frequencies, not bending natural frequencies:

$$f_{\text{spin}} = \frac{n}{60}, \quad f_{\text{tooth}} = 6 \frac{n}{60}, \quad \left. \frac{f_{\text{spin}}}{f_R} \right|_{3000 \text{ rpm}} = \frac{50}{336.78244} = 0.14846. \quad (62)$$

The ratio 0.14846 alone does not specify a percentage correction. In this model the gyroscopic split also depends on g/m , through $\varepsilon = (g/2m)(\Omega/\omega_0)$. A stationary natural frequency alone is insufficient to locate a speed-dependent synchronous critical speed.^[14]

A Campbell view of this diagnostic, not of the full machine

A Campbell diagram compares frequency branches with speed-related excitation lines. In addition to the 1X line, the six-flute example needs a 6X line. A crossing identifies a possible frequency coincidence; the resulting response also needs modal forcing and damping.^[11]



The FE marker is the report's single 3000 rpm result, not a computed FE branch. Its offset from the reduced curves includes geometry, restraint, retained-subassembly, and spin-scope differences. The two reduced branches are nearly superposed at this scale; page 16 enlarges their splitting.

Do not conclude that 50 Hz being far below the bending mode makes cutting excitation harmless. The 300 Hz tooth-passing fundamental is only 7.17 Hz below the reported 307.17 Hz FE mode. Conversely, proximity alone does not prove a large response. Higher harmonics, other modes, and regenerative cutting effects remain outside this diagnostic.

17 / MATCHED FE COMPARISON

Separate structural discrepancy from speed effects

The present hand-versus-FE difference mixes several changes. A paired FE extraction would distinguish the stationary structural-model discrepancy from the speed effect in the *same* assembly:

$$\Delta f_{\text{structure}} = f_{\text{hand}}(0) - f_{\text{FE}}(0), \quad (63)$$

$$\Delta f_{\text{speed},\pm} = f_{\text{FE},\pm}(3000) - f_{\text{FE}}(0). \quad (64)$$

Each subtraction must use the corresponding bending family and the same reference frame. No such matched zero-speed FE result was supplied; these quantities are not evaluated here.

A controlled comparison to perform in the original project

1. **Recover the actual model.** Record geometry, material assignments, bearing/joint scopes, contact treatment, spin assignment, and any prescribed physical damping. In particular, resolve the stationary and rotating parts of the rotary union.
2. **Extract at zero and 3000 rpm, then sweep the intervening speeds.** Keep the underlying geometry, supports, contacts, and damping specification unchanged. State whether the equilibrium/prestress is recomputed at each speed. A speed sweep is the appropriate way to inspect frequency evolution.^[14]
3. **Track the bending family, not merely “mode 1.”** Compare complex mode shapes, whirl direction, and subspace continuity near the zero-speed pair. Individual vectors can change basis inside a degenerate pair. Compare outputs in the same stationary frame.^[11]
4. **Isolate the requested effect.** A total speed-effect check may include speed-dependent prestress and stiffness. A gyro-only check must instead hold mass and stiffness fixed while varying the gyroscopic term. Report the distinction rather than attributing every change to gyroscopy.

Centrifugal prestress is not the gyroscopic term

This extension adds m_θ and ΩG_1 , while holding K fixed. It does not compute centrifugal stress or its effect on tangent stiffness. Rotating-frame spin-softening terms, stationary-frame gyroscopic terms, and damping-related rotational terms are not interchangeable; their representation depends on the selected frame.^[12] Do not add a guessed $-m\Omega^2$ stiffness to this stationary-frame result.

“Damped solver” does not quantify dissipative damping

The report records a damped modal solver and Coriolis effects, but that setting alone supplies no measured damping ratio. ANSYS Mechanical’s 2026 R1 documentation enables the modal Coriolis options only when the Damped property is set to Yes.^[13] This explains why the label is not evidence that substantial physical damping caused the hand–FE frequency difference; it does not audit the original 2025 solver deck.

What has been checked: a fixed-shape, fixed-stiffness gyroscopic extension of the assumed overhang gives approximately $\pm 0.208\%$ at 3000 rpm.

What remains unchecked: the full spindle’s speed dependence, actual spinning-component inertias, finite bearing restraint, centrifugal prestress, damping, and cutting-response amplitude. The added calculation supports a baseline simplification; it does not certify resonance avoidance.

18 / INTERVIEW EXPLANATION

The story to explain at a whiteboard

A historically accurate explanation, including the rotation check

“The original ANSYS mode concentrated bending in the motor-end overhang. I later modeled that region as a stepped beam with an ideal lateral clamp. A unit transverse force gave its compliance and an effective stiffness of about 3.08 MN/m. Weighting the distributed mass by the deflected shape gave 0.688 kg, so Rayleigh’s method predicted about 337 Hz. A two-station version gave 334 Hz, about 8.6 percent above the reported 307 Hz. I then checked a separate two-plane extension of the original static shape. Cross-sectional rotary inertia lowered the zero-speed result by about 0.7 percent; gyroscopic coupling at 3000 rpm shifted it by about two-tenths of a percent in either direction. That supports omitting gyroscopy from the baseline explanation within the assumed beam. The co-rotation assumption does not match every real component, so I would verify it with matched FE speed cases. Neither calculation validates the image-estimated geometry or the actual bearing restraint.”

Why a cantilever rather than simple supports?

The retained motion is beyond the motor-side bearing. The recorded cylindrical-joint idealization restrains transverse rotations as well as translations. The local clamp reflects that limit, not an assertion that every rolling bearing is rigid against tilt. The separate two-radial-support example is on page 12.

Why is effective mass smaller than the assembly mass?

Kinetic energy weights mass by the square of its modal displacement. Heavy material near a nearly stationary root can contribute little. Here “effective mass” means tip-normalized generalized mass, not ANSYS’s base-excitation participation metric. Gyroscopic inertia instead depends on the slope-weighted polar inertia.

Are the whirl frequencies your first and second bending modes?

No. The 333.73/335.13 Hz pair uses one spatial shape in perpendicular planes. The original two-station model has distinct local bending modes at about 334 and 1243 Hz. The report’s second system frequency, 523.67 Hz, has no supplied mode shape and cannot be identified with that local second mode.

What did the hand calculation establish about the redesign?

A 22 mm root translation raises the original first Ritz frequency by about 7.4%, not the report’s 24.8%. It explains the direction of the support-location effect, while a spaced bearing pair can also resist tilt. Actual bearing count, spacing, and joint scopes are needed to reconstruct the reported redesign.

Preserve the chronology. The team project and ANSYS results are from spring 2025. The hand calculations and rotation check were added in September 2026. Do not present them as analyses that drove the original design, or a simulated frequency change as a measured post-installation vibration reduction.

19 / DYNAMICS LIMITS

Frequency relocation is not vibration amplitude

Spindle speed and tooth-passing frequency are different

The report specifies up to 3000 rpm and six flutes.^[1, pp. 8–9] Therefore

$$f_{\text{spin,max}} = \frac{3000}{60} = 50 \text{ Hz}, \quad f_{\text{tooth,max}} = 6 \frac{3000}{60} = 300 \text{ Hz}. \quad (65)$$

The 307 Hz mode lies near the top of the *six-tooth fundamental* range, not near the once-per-revolution frequency at 3000 rpm. Equating a constant 307.17 Hz natural frequency with 1X would give about 18,430 rpm, outside that operating range. This conversion is not a rotordynamic critical-speed prediction.

Raising a mode above 300 Hz separates it from the maximum fundamental tooth-passing frequency, but does not remove higher harmonics, broadband excitation, or regenerative cutting instability. The unloaded local modal model does not establish chatter-free operation.

The rotation check does not close the original FE comparison

The baseline remains nonrotating. The new extension (pages 13–18) checks rotary inertia and gyroscopic coupling, not the full rotating assembly. A broader linearized description can include

$$M\ddot{q} + (C_d + \Omega G_1)\dot{q} + K(\Omega)q = f(t). \quad (66)$$

The reported FE case uses 3000 rpm, excludes bearings and the rotary union from assigned rotation, and selects a damped modal solver.^[1, p. 28] Its spin scope and stiffness treatment have not been matched here. The new $\pm 0.208\%$ shift is not a total rotation-error bound, and the solver label alone does not establish physical damping.^[13] Use the controlled comparison on page 18 before attributing the hand–FE difference to rotation.

A modal displacement legend does not predict operating motion

In an eigenproblem, any multiple of an eigenvector is still an eigenvector. The 25.056 mm maximum shown in the original first-mode image is therefore not a physical vibration amplitude. ANSYS likewise cautions that modal displacement/stress magnitudes are not actual response values.^[5]

For a single forced mode,

$$|q(\omega)| = \frac{|F_{\text{modal}}|/k_{\text{modal}}}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \quad r = \frac{\omega}{\omega_n}. \quad (67)$$

Amplitude needs the forcing, damping, and relevant mode shape. A frequency shift alone does not demonstrate a reduction in measured RMS velocity, meet an ISO amplitude limit, or prove increased damping.

Keep unloaded and cutting claims separate. The original report states that its loaded simulation was not successfully validated. Its impact tests were inconclusive, and some loaded long-tool recordings saturated.^[1, pp. 19, 24–25] This hand note does not repair those evidence gaps or derive the nonlinear cutting response from an unloaded natural frequency.

20 / BASELINE REPRODUCIBILITY

Check the reduction separately from the real machine

An internal numerical check of the same assumed beam

A small, independent Euler–Bernoulli beam discretization with consistent distributed mass was assembled in Python. It uses the same five sections, ring mass, and clamped root, without cross-sectional rotary inertia or spin. This is a verification of the reduced calculation, **not a new ANSYS analysis or physical validation**.

Discretization / approximation	Local mode 1 (Hz)	Local mode 2 (Hz)
6 beam elements	331.16	1228.14
12 beam elements	331.15	1227.25
24 beam elements	331.15	1227.20
One-coordinate Rayleigh approximation	336.78	Not retained
Two-coordinate Ritz approximation	333.70	1242.93

Relative to the 24-element check, the two-coordinate frequencies are 0.77% and 1.28% higher. The single-coordinate first estimate is 1.70% higher. The positive differences are consistent with restricted Rayleigh–Ritz trial spaces for this self-adjoint beam problem; they are not upper bounds relative to the actual assembly or original rotating FE model.

What the calculation script verifies

`verify_spindle.py` recomputes section stiffnesses, static flexibility, generalized masses, both eigenvalues, the 22 mm shift, sensitivities, and all comparison percentages. It checks symmetry, positive mass, station interpolation, stiffness from strain energy, eigenvector mass orthogonality, and eigen-equation residuals. Four-point Gaussian integration on each polynomial interval exactly integrates the mass terms in exact arithmetic; floating-point roundoff remains.

Original calculation files	Purpose
<code>spindle_hand_calculations.tex</code>	Typeset report and vector schematics.
<code>verify_spindle.py</code>	Standalone calculation and internal beam check; edit section inputs at the top.
<code>data/results.json</code>	Full-precision inputs, matrices, and numerical results.
<code>data/mode_shapes.dat</code>	Computed curves used by the LaTeX plot.
<code>assets/</code> and <code>README.md</code>	Original extracted figures, provenance, limitations, and rebuild instructions.

What would turn this into a stronger model-to-model check?

Replace the silhouette estimates with a CAD station table, including bores and coaxial parts. Record each component’s mass and connection path. Recover the baseline and modified joint locations/scopes, exact bearing count and spacing, and modal speed settings. Then compare corresponding zero-speed mode shapes as well as frequencies. Finite bearing stiffness and housing motion should be added only with identified parameters or explicitly labeled sensitivity cases.

Baseline conclusion: the hand calculation explains a plausible motor-end bending mechanism and lands within about 9% of the reported first frequency. Its mathematical reduction is much more tightly checked than its physical inputs. The rotation-specific checks on the next page preserve that distinction.

21 / ROTORDYNAMIC REPRODUCIBILITY

Check the new arithmetic and its interpretation

`verify_rotordynamics.py` imports the original section definitions without changing them. It reconstructs the static deflection section by section, integrates the polynomials directly, and compares them with the original influence-function shape and separate Gaussian integration.

Independent algebraic check of the same reduced system

Let $\tau = \omega_0 t$ and $\mathbf{z} = [\mathbf{q}; d\mathbf{q}/d\tau]$. The nondimensional four-state eigenproblem is

$$\frac{d\mathbf{z}}{d\tau} = \mathbf{A}\mathbf{z}, \quad \mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I}_2 \\ -\mathbf{I}_2 & -\frac{\Omega}{m\omega_0}\mathbf{G}_1 \end{bmatrix}. \quad (68)$$

Its eigenvalues occur as imaginary conjugate pairs. Their positive imaginary parts, multiplied by $\omega_0/(2\pi)$, reproduce the closed-form frequencies:

$$f_- = 333.73470894 \text{ Hz}, \quad f_+ = 335.12681598 \text{ Hz}. \quad (69)$$

This verifies the implementation of the two-plane reduction, not the physical adequacy of the beam or the original ANSYS model.

Check	Result of the supplied calculation
Static shape and slope	Polynomial recurrence agrees with the original influence-function shape and its derivative.
Translational mass and stiffness	Original $m_t = 0.6884325563 \text{ kg}$ and $k = 3.0826201615 \text{ MN/m}$ reproduced.
Rotary integral	Polynomial and Gaussian integration agree within floating-point precision.
Gyroscopic structure	$\mathbf{G}_1^T = -\mathbf{G}_1$ and $g = 2m_\theta$ for this co-rotating circular-section case.
Zero and reversed spin	Branches coincide at $\Omega = 0$ and exchange under $\Omega \mapsto -\Omega$.
Frequency identities	$\omega_+\omega_- = k/m$ and $\omega_+ - \omega_- = g\Omega/m$ reproduced.
Four-state eigensolution	Relative eigen-equation residual below 10^{-12} ; closed-form agreement better than 10^{-10} Hz .

Additional editable outputs

`data/rotordynamic_results.json` contains section slopes and inertia contributions, the four frequency cases, percentage denominators, and numerical checks. `data/whirl_branches.dat` supplies the plotted 0–3000 rpm branches and 1X/6X excitation lines. `data/rotordynamic_verification_log.txt` records the calculation output.

Do not confuse arithmetic reproduction with model validation. These checks keep the same assumed shape and ideal support. They do not assess sensitivity of the gyroscopic correction to a different bearing restraint, actual component inertias, additional spatial modes, or the missing centrifugal-prestress terms. No rotating beam-mesh convergence study or new ANSYS solve is claimed.

22 / ORIGINAL REFERENCES AND EVIDENCE

Where the inputs and comparisons come from

[1] **Primary project record.** Z. Balakrishnan, R. VandeWiele, E. Mahmudi, L. Wydra, J. Lee, and J. Frank. *Spindle Vibration and Deflection Reduction*. University of Illinois Urbana–Champaign, Spring 2025 ME 470 final report, 9 May 2025. User-supplied file: SVRSP25FinalReport–2.pdf. PDF page numbers are one greater than the report’s printed numbers after the title page.

PDF pages	Specific evidence used in this note
10, 15–16	Speed/flutes; fixed and floating bearings; shaft, gear, drawbar, rotary union, C-ring anatomy.
20, 23, 25	Inconclusive impact tests; unloaded band near 310 Hz; saturated loaded data.
26–30	Material assignments and Table 2; cylinder approximation; Figure 13 mesh; joint, grounding, rotation, and solver descriptions.
31–32	Frequency tuning narrative; Figure 18 at 307.17 Hz; mesh results 305–307 Hz.
34–35, 42	22 mm relocation; inconsistent bearing counts; Figure 21 at 383.27 Hz; reported modification series.
37	Table 5: baseline 307.17 / 523.67 Hz and steel-grade comparisons.

[2] **Project portfolio.** J. Lee. *Spindle Vibration & Deflection Reduction*. Accessed 7 September 2026. [Project page](#). Used for project context; the more detailed supplied report controls numerical and setup interpretation. The page’s broader claims of mode-shape validation and safe separation from cutting excitation are not independently established by this note.

[3] **Joint degrees of freedom.** ANSYS. *Multibody Analysis Guide*, release 2024 R2, §2.3, “Connecting Multibody Components with Joint Elements.” [Official documentation](#). In particular, the cylindrical-joint constraint list. Consulted 7 September 2026; the original project deck was not available for an element-level audit.

[4] **Attachment behavior.** ANSYS. *Mechanical User’s Guide*, release 2024 R2, §12.2, “Geometry Behaviors.” [Official documentation](#). Deformable versus rigid remote geometry attachment. Consulted 7 September 2026.

[5] **Modal interpretation.** ANSYS. *Mechanical User’s Guide*, release 2025 R2, §5.6.5, “Modal Analysis.” [Official documentation](#). Modal scaling, joint constraints, damped/rotating analyses, and prestress discussion. Consulted 7 September 2026. Referencing these instructions does not confirm the original model’s actual solver implementation.

[6] **Simple rotor models.** J. E. T. Penny, M. I. Friswell, A. W. Lees, and S. D. Garvey. *A simple but versatile rotor model*. C623/033/2004, 2004, opening discussion on printed p. 269. [Author-hosted paper](#). Supports the Jeffcott-model context; the present stepped-overhang derivation is a separate reduction.

[7] **Rayleigh energy method.** University of Alberta, Engineering at Alberta Courses. *Vibrations and Sound: Application to Lateral Vibrations of Beams*. [University course resource](#). Method background; all numerical sections, matrices, and modal curves in this note were computed for the explicitly assumed geometry.

[8] **Methodological precedent.** *Pigjaw push-out fixture: Hand calculations, retrospective PLA FEA, and model comparison*, CALC-01 Rev. C, 7 September 2026. User-supplied file: *Pigjaw_Fixture_Material_Hand_Calculations_Rev_C.pdf*. Used as the model for separating sourced inputs, assumptions, model discrepancies, and retrospective project history; no spindle dimensions or properties are taken from it.

Evidence boundary. This package contains no original spindle CAD, solver archive, raw modal test records, measured bearing stiffness, or as-built verification. Extracted report figures are preserved as illustrations, not newly generated simulations. The calculation is an engineering explanation and screening model, not a load rating, commissioning assessment, or certification of machine performance.

23 / ADDED REFERENCES AND REVISION

The rotordynamic addition and its evidence

[9] **Beam-rotor inertia formulation.** ANSYS. *Mechanical APDL Theory Reference*, release 2024 R2, §14.4, “Rotating Structures,” especially §14.4.2.1–2 on gyroscopic matrices and lumped/beam kinetic energy. [Official documentation](#). Supports the physical formulation; the integrals and numerical values are evaluated for this note’s assumed sections.

[10] **General rotor equation.** ANSYS. *Rotordynamic Analysis Guide*, release 2024 R2, §1.1, “General Dynamics Equations.” [Official documentation](#). Distinguishes structural mass, stiffness, damping, gyroscopic, and other rotational contributions.

[11] **Whirl, Campbell diagrams, and reference frame.** ANSYS. *Rotordynamic Analysis Guide*, release 2024 R2, §1.2, “Some Basic Terminology.” [Official documentation](#). Whirl direction, speed-dependent modes, excitation intersections, and stationary-frame frequency interpretation. The note’s sign convention and one-shape algebra are stated explicitly on page 15.

[12] **Distinct rotational terms.** ANSYS. *Advanced Analysis Guide*, release 2025 R1, §5.1, “Understanding Rotating Structure Dynamics.” [Official documentation](#). Clarifies gyroscopic/Coriolis, spin-softening, rotating-damping terms, and reference-frame dependence. No centrifugal-stress stiffness correction is calculated in this note.

[13] **Modal Coriolis controls.** ANSYS. *Mechanical User’s Guide*, release 2026 R1, “Rotordynamics Controls.” [Official documentation](#). Its modal settings table requires Damped = Yes for the Coriolis options. This later documentation is not evidence of a measured damping ratio or an audit of the original 2025 model.

[14] **Speed-sweep workflow.** P. Srivastava. *Analyzing Critical Speeds with the Rotor Bearing System Simulator*. COMSOL, 25 May 2017. [Official vendor article](#). Supports using speed-dependent eigenfrequencies rather than only a stationary frequency to assess synchronous critical speed.

References [9]–[14] were checked on 13 September 2026. References [1]–[8] and their recorded consultation dates are carried forward from Revision A. The user’s supplied rotordynamic analysis is the basis of this addition; numerical values were recomputed from the preserved section inputs, and linked formulation/solver statements were checked against the official sources above.

Revision history

Revision	Date	Change
A	7 September 2026	Stepped-beam baseline, two-station Ritz model, reported FE comparison, support relocation, sensitivities, and numerical beam check.
B	13 September 2026	Adds two-plane rotary-inertia/gyroscopic sensitivity, stationary-frame whirl branches, Campbell view, matched-FE plan, reproducibility checks, and revised summary/interview/limitations. No new ANSYS solve.

Notation-only change: the original gear-center station $g = 0.095\text{ m}$ is written s_g so that g can denote the generalized gyroscopic coefficient. Baseline geometry, material properties, and all original numerical scenarios are retained. The original source bundle and PDF are preserved in the companion package’s `evidence/revision_a/` folder.

Revised conclusion. The nonrotating beam remains a useful explanation of the motor-end bending mechanism. A same-shape gyroscopic check gives approximately $\pm 0.208\%$ at 3000 rpm under the declared co-rotation assumption. This justifies the baseline simplification within that surrogate, not precision operating-speed or resonance-avoidance claims for the complete spindle.